

Infrastructure Memo: Subscriptions for Fute Portal

What to pay for, what to keep free, and why. Sized for 1,000 to 2,000 daily logins in production.

Prepared 2026-08-14. Stack: Vercel + Firebase. Scale: 1 to 2 thousand daily users.

Recommendation

Stay on the current stack. Upgrade Firebase to its "Blaze" (pay-as-you-go) plan and Vercel to its "Pro" plan. Both are required at this scale, not optional extras. Everything else on this page is free-tier and stays that way.

1. Subscription decisions

SERVICE	PLAN	COST	WHEN	WHY
Firebase Firestore + Auth (the database and login system)	Blaze (pay-as-you-go)	About ₹800 to ₹4,000/month	Now	The free tier caps at 50,000 database reads a day. At 1 to 2 thousand active users this gets used up within hours, not days. The app will start failing without this.
Vercel (backend + frontend hosting, meaning where the website and its server-side code run)	Pro	\$20/month	Now	The free "Hobby" plan's terms restrict it to personal, non-commercial use. A company app serving daily active users needs the Pro plan regardless of how much traffic it gets.
Sentry (error tracking)	Developer (free)	₹0	Later	5,000 errors a month, free, comfortably covers this scale. Add it when you're ready to wire it in. No rush to pay.
UptimeRobot (uptime alerts, which tell you if the site goes down)	Free	₹0	Later	50 monitors, checks every 5 minutes, free forever at this scale. No paid tier needed.
Custom domain	Registrar of choice	About \$12/year	Optional	This is cosmetic only. The Vercel-issued web address works fine as-is. Add a custom domain only if you want a branded link.

Realistic monthly total (the "Now" items): roughly ₹2,500 to ₹7,300.

The range is wide because the Firebase cost depends entirely on the read pattern described below. The same feature can cost 5 times more or less depending on one implementation choice.

2. Why the Firebase cost has a range

If polling stays as-is: about ₹4,000 to ₹4,500/month. Tickets and Approvals currently poll (automatically check for updates) every 20 seconds per user, regardless of whether anything actually changed. At 1,000+ people using the app at once, this alone generates millions of billable database reads a day.

If switched to listeners: about ₹800 to ₹1,200/month. Firestore's "onSnapshot" real-time listener (a way of watching for changes that only reports back when data actually changes) only bills a read when data actually changes. Same feature, same number of users, but a fraction of the cost, and updates arrive instantly instead of within 20 seconds.

3. What NOT to add

Railway, Render, or a dedicated virtual server: none of these buy you anything here. They exist for things like persistent connections (websockets) and long-running background jobs, and this backend has neither. Adding one would mean paying for a second hosting bill and doing a migration, for zero functional gain.

Fute Portal, internal infrastructure memo. Figures are estimates based on published Vercel and Firebase pricing. Actual Firebase cost should be confirmed against the Blaze calculator once traffic is live.